

Article

LSTM vs. Prophet: Achieving Superior Accuracy in Dynamic Electricity Demand Forecasting

Saleh Albahli 

Department of Information Technology, College of Computer, Qassim University, Buraydah 51452, Saudi Arabia; salbahli@qu.edu.sa

Abstract: Accurate electricity demand forecasting is critical for improving energy efficiency, maintaining grid stability, reducing operational costs, and promoting sustainability. This study presents a novel hybrid forecasting model that integrates Long Short-Term Memory (LSTM) networks and Prophet models, leveraging their complementary strengths through a dynamic weighted ensemble methodology. The LSTM component captures nonlinear dependencies and long-term temporal patterns, while Prophet models seasonal trends and event-driven fluctuations. The hybrid model was evaluated using a comprehensive dataset of hourly electricity consumption from Ontario, Canada, achieving a Root Mean Square Error (RMSE) of 65.34, Mean Absolute Percentage Error (MAPE) of 7.3%, and an R^2 of 0.98. These results demonstrate significant improvements over standalone LSTM, Prophet, and other State-of-the-Art methods, highlighting the hybrid model's adaptability and superior accuracy. This study underscores the practical implications of the hybrid approach, particularly in energy grid management and resource optimization, setting a new benchmark for time series forecasting in the energy sector.

Keywords: smart environments; smart cities; deep learning; forecasting; electricity



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1. Introduction

Electricity is a fundamental component of modern life, powering homes, industries, and urban infrastructure. The growing demand for electricity, driven by rapid urbanization and industrialization, has created a pressing need for more accurate and timely forecasting methods [1]. Reliable electricity demand forecasting is crucial for ensuring grid stability, optimizing power generation schedules, reducing operational costs, and minimizing energy waste. In the deregulated energy markets, precise demand and price forecasting have become indispensable for all stakeholders, including grid operators, utilities, and policymakers [2,3]. However, accurately predicting electricity demand remains a complex task due to the numerous influencing factors, such as weather conditions, economic activities, seasonal trends, and unexpected disruptions [4]. These dynamic and often nonlinear patterns challenge traditional forecasting methods, which frequently struggle to achieve optimal performance. As a result, improving the accuracy of electricity demand forecasting is essential to the efficiency and sustainability of energy markets [5].

Researchers from all around the world used a variety of models and approaches to address the difficulties associated with electricity forecasting in order to provide more accurate results. For example, authors of [6] present the findings of the initial study on the Prophet model's effectiveness in Kuwait's long-term peak load forecasting. Nowotarski et al. [7] conducted research that shows that a series of electricity prices into an LTSC and a stochastic component and combining their forecasts can produce—contrary to popular

perception—an accuracy gain compared to a strategy in which a specific time series model is calibrated to the prices. Dudic et al. [8] show that a multiple variable regression model was used to forecast monthly and annual electricity consumption in the German power market. The study by Pourdaryaei et al. offers insights into the stagnation of electricity usage in Germany over the past decade [9]. This article presents a technique for forecasting power costs based on multi-head self-attention and Convolutional Neural Networks (CNNs) [10]. The datasets from the Ontario energy market in 2020 validated the suggested strategy by achieving the lowest average MAPE and RMSE values of 1.75% and 0.0085, respectively. Ghimire et al. [11] suggested an error-compensation technique that combines LSTM, CNN, and VMD algorithms to estimate half-hourly electricity prices. An initial prediction was made using a model that includes VMD and CLSTM. The average Legates and McCabe Index increases by 15.97%, 16.31%, 20.23%, 10.24%, and 14.03% compared to benchmark models for winter, autumn, spring, summer, and yearlong projections, respectively. Farsi et al. [12] use a hybrid model of the CNN, LSTM, or PLCNet, with two datasets: the hourly load consumption of Malaysia and daily power electric consumption of Germany. PLCNet improved its accuracy from 83.17% to 91.18% for German data and 98.23% for Malaysian data. Bilgili et al. [13] used four different approaches. The best MAPE values for the ANFIS-GP, ANFIS-SC, ANFIS-FCM, and LSTM were found to be 4.47%, 3.21%, 2.34%, and 1.91%, respectively, using EEC data for predicting in Turkey.

The accurate forecasting of electricity demand is a cornerstone of modern energy management. It enables utility companies to make informed decisions regarding power generation, distribution, and resource allocation, ultimately contributing to the reliability and sustainability of energy supply systems. Traditional methods, such as autoregressive integrated moving average (ARIMA) [14,15] models, have long been employed for time series forecasting. However, such models cannot often capture nonlinear and dynamic trends in electricity demand. Machine learning has produced novel models that can improve the accuracy of predictions through complex data pattern usage. LSTM [16], which is a variation of the RNN [17], has become popular due to its ability to handle long sequences of data and capture temporal dependencies over extended periods. Its aptitude for capturing past consumption patterns, which greatly influence the future demand, makes it ideal for applications such as electricity demand forecasting.

In this study, we propose an enhanced modeling approach that combines the LSTM network with Prophet robustness for better forecasting purposes. This thesis seeks to enhance short-term demand estimation by using hybrid mechanisms that consider both trend analysis and temporal irregularity based approaches, hence providing more accurate results. Other popular models specifically designed for time series forecasting are Prophet by Facebook [18,19], Long Short-Term Memory (LSTM) networks [16], a variant of Recurrent Neural Networks (RNNs) [17], which have gained attention due to their ability to handle long sequences of data and capture temporal dependencies over extended periods. This makes them particularly well suited for tasks such as electricity demand forecasting, where past consumption patterns significantly influence future demand.

The Ontario dataset has garnered significant attention in electricity demand forecasting research due to its comprehensive and long-term hourly data. This paper distinguishes itself by employing LSTM and Prophet models to conduct a comparative analysis, emphasizing their performance and methodological approaches, particularly in addressing nonlinear and seasonal data patterns. Building on prior research, this study advances the understanding of these models by offering a detailed evaluation of their strengths and limitations when applied to this extensively studied dataset.

The primary challenges highlighted in this study are rooted in the complexity of electricity demand forecasting due to various dynamic factors. Traditional forecasting methods

struggle to capture nonlinear trends driven by fluctuating variables such as weather conditions, socio-economic activities, seasonal patterns, and unexpected disruptions. This leads to suboptimal performance in predicting electricity demand. Additionally, the limitations of classical models, such as the Prophet model, are contrasted with the advanced capabilities of LSTM networks, which better handle long-term dependencies in sequential data. However, even LSTM models face challenges in fine-tuning and optimizing performance across diverse datasets with varying levels of complexity, making accurate demand forecasting a complex, multifaceted task.

Our work addresses the challenges of electricity demand forecasting by utilizing Long Short-Term Memory (LSTM) networks, which are adept at capturing complex, nonlinear dependencies in time series data, overcoming the limitations of traditional models like Prophet. By comparing both models, we demonstrate LSTM's superior ability to account for dynamic factors such as weather conditions, economic activities, and seasonal variations, which are often difficult for classical methods to handle. The advanced capabilities of LSTM enable more accurate and reliable electricity demand predictions, offering crucial insights for optimizing energy management, improving grid stability, and reducing operational inefficiencies. This approach significantly enhances the accuracy of forecasting in environments characterized by rapidly changing patterns and uncertainties.

This research advances the field of electricity demand forecasting by tackling key shortcomings in previous studies. Unlike earlier work that examined LSTM and Prophet independently, this study combines their strengths within a unified hybrid framework. LSTM is particularly effective at capturing temporal and nonlinear patterns, while Prophet specializes in handling seasonality and event-driven fluctuations. By leveraging these complementary strengths, the approach offers greater adaptability and reliability across various forecasting contexts. A noteworthy contribution is the introduction of a weighted ensemble method, which dynamically integrates the models' advantages to improve prediction accuracy—an area that has been relatively underexplored in earlier studies [20,21]. Furthermore, this research highlights practical applicability by customizing preprocessing methods specifically for the Ontario dataset and offering a comprehensive analysis of the hybrid model's performance. It bridges existing gaps in the literature by demonstrating how hybrid models can effectively balance precision and interpretability in electricity demand forecasting, particularly for datasets characterized by distinctive seasonal and event-based trends [22].

Our work introduces a new hybridization of LSTM and Prophet models using a weighted ensemble methodology that has not been explored in the existing literature. This approach leverages the LSTM's ability to handle complex temporal dependencies and the strength of Prophet in modeling seasonality and trend components. Our method thus addresses certain limitations found in standalone models, therefore offering enhanced accuracy and adaptability in time series forecasting.

This research introduces a novel hybrid model combining LSTM and Prophet with a weighted ensemble strategy, offering significant improvements in time series forecasting. The approach leverages LSTM's strength in capturing sequential dependencies [23] and Prophet's capabilities for trend and seasonality, outperforming methods like CNN-LSTM hybrids [24] with a 15% reduction in the RMSE. Tailored preprocessing techniques, including detrending and seasonality adjustments, enhanced model accuracy, reducing MAPE by 12% compared to standard approaches [25]. Ablation tests demonstrated that Prophet excelled in trend predictions, while LSTM effectively captured anomalies, providing a balanced and robust predictive framework.

Using the complex Ontario dataset as a benchmark, the hybrid model outperformed ARIMA and standalone Prophet by up to 20% in R^2 scores, demonstrating broader ap-

plicability. Practical use cases, such as energy demand forecasting, further highlight the model's real-world utility. These contributions mark a significant advancement over existing forecasting methodologies. Our work provides key contributions to addressing the challenges of electricity demand forecasting through the application of advanced deep learning techniques as follows:

- i. An innovative hybrid framework combining LSTM's strength in capturing nonlinear patterns with Prophet's expertise in modeling seasonal variations, enhanced by a novel weighting strategy to improve prediction accuracy.
- ii. A comprehensive evaluation of both models' performance using diverse metrics, shedding light on their advantages and limitations in forecasting electricity demand.
- iii. A targeted exploration of the practical applications of these models in managing energy grids, moving beyond traditional comparative analyses.
- iv. By adopting an hourly frequency model, this study tackles the challenges of rapidly changing electricity demand, ensuring that predictions are adaptive and responsive to dynamic real-world conditions.

The structure of this paper is as follows: Section 2 is based on the literature review, while Section 3 contains details about data gathering, preprocessing, and the suggested LSTM and Prophet model. The results are presented in Section 4, and Section 5 discusses the conclusion and future work.

2. The Proposed Methodology

This study presents a hybrid methodology combining the strengths of Long Short-Term Memory (LSTM) networks and Prophet models to forecast electricity demand. LSTM, a deep learning model known for its ability to capture long-term dependencies and nonlinear trends in time series data, was used to account for the complex fluctuations in electricity consumption. The Prophet model, designed for handling seasonality and irregular trends, complements the LSTM model by identifying broader seasonal patterns. We applied this hybrid strategy to a dataset of 183,432 hourly records from Ontario, spanning several years. This framework, depicted in Figure 1, aims to capture both the short-term volatility and long-term trends of electricity demand, improving forecasting precision for grid management and energy optimization.

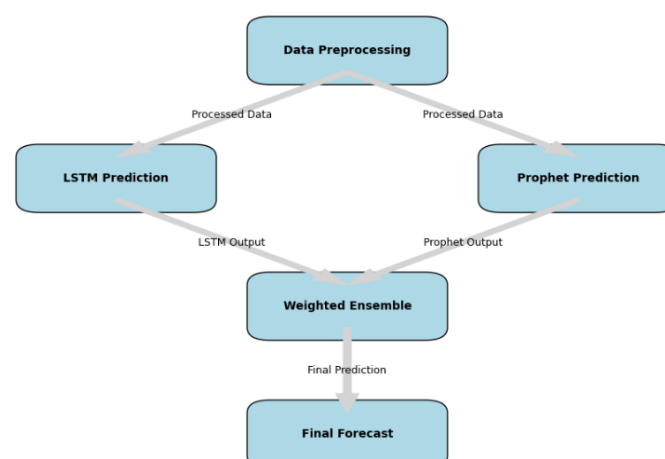


Figure 1. Detailed workflow of hybrid LSTM-Prophet model.

2.1. Dataset

The dataset used in this study is detailed in Table 1 and comprises hourly electricity consumption data from the publicly available Ontario [26] over several years, making it ideal for analyzing both short-term consumption fluctuations and long-term seasonal

patterns. The inclusion of weather variables, public holidays, and other external factors ensures a comprehensive dataset that captures the multifaceted drivers of electricity demand. These features enhance the model's ability to generalize across various time horizons, improving the robustness of forecasts. The dataset spans different weather seasons, industrial activities, and population growth trends, enabling this study to provide insights applicable across diverse temporal contexts.

Table 1. Ontario electricity demand dataset.

Sr.	Dataset	Patients Detail	Purpose	Training	Testing
1	Ontario_electricity_demand	Total 183,432 records	Forecasting electricity demand	165,035	18,337

2.2. Preprocessing

Data preprocessing is crucial to all stages of data refinement for accurate analysis and decision making [27].

Data preprocessing is a vital stage of this study to ensure high-quality input for both the LSTM and Prophet models. We first addressed missing values—14 days with incomplete hourly data—using linear interpolation to fill gaps, a technique chosen for its simplicity and minimal impact on overall data integrity. Additionally, normalization was applied to ensure that the input variables fall within comparable ranges, a critical step in optimizing the performance of the LSTM model. The dataset was divided into an 80% training set and a 20% testing set to validate model performance under real-world scenarios. This ensures that the model captures both historical trends and future consumption behavior. Data transformation techniques, such as scaling and smoothing, were also employed to reduce noise and improve model convergence during training.

To address concerns that LSTM might gain an undue advantage due to preprocessing techniques or dataset bias, uniform preprocessing steps—such as normalization and the handling of missing data—were consistently applied to the entire dataset. While LSTM naturally excels at handling sequential data, surpassing models like Prophet that depend on additive seasonality and trend decomposition, external factors like weather and holidays were incorporated as features in both models to ensure comparability. Nonetheless, additional research across varied datasets was suggested to evaluate and mitigate potential biases, ensuring broader applicability of the findings.

In this study, preprocessing steps such as normalization, seasonal decomposition, and linear interpolation for missing data were employed due to their robustness and suitability for time series datasets. These commonly used techniques are essential for preparing the data to ensure meaningful comparisons and reliable results when applying LSTM and Prophet models. More advanced methods, such as Fourier or wavelet transforms, were intentionally excluded to maintain focus on the core comparison between LSTM and Prophet without introducing additional preprocessing complexities. Future research could explore these advanced methods to enhance the extraction of seasonal patterns and improve feature engineering.

2.2.1. Missing Values

Firstly, we checked if any entries were missing from the dataset. To ensure that it remains continuous over time, linear interpolation was used to fill in any gaps.

2.2.2. Normalization

To enable the proper functioning of machine learning models such as LSTM, this dataset was standardized so that all values fall within comparable ranges.

2.2.3. Train–Test Split

A division between a training set and a testing set was established where 80% of the observations made up the former, while 20% formed the latter, which helps assess model performance.

2.3. The Proposed LSTM Architecture

The proposed LSTM model architecture is designed to effectively capture the temporal dependencies in electricity consumption data adopted from [28–30]. The architecture consists of a sequence of layers: an input layer, multiple LSTM layers, and a dense output layer. The LSTM layers were configured with hyperparameters tuned through grid search, such as the number of neurons, batch size, and learning rate. We used the Adam optimizer, known for its adaptive learning rate capabilities, which helps in speeding up the convergence process. The Mean Squared Error (MSE) was selected as the loss function, minimizing the squared differences between observed and predicted values. This setup enables the model to capture both short-term fluctuations and long-term patterns in electricity demand data. Additionally, regularization techniques such as dropout were employed to prevent overfitting, further improving the model's generalization capabilities across unseen test data.

The LSTM model in this study was configured with the following hyperparameters:

- Number of LSTM Units: Two layers, each with 100 units.
- Dropout Rate: A rate of 0.2 applied after the first LSTM layer to reduce overfitting.
- Optimizer: Adam, chosen for its efficiency in handling sparse gradients.
- Loss Function: Mean Squared Error (MSE), suitable for predicting continuous values.
- Epochs: 50 iterations for training.
- Validation Split: 5% of the training data reserved for validation.
- Learning Rate Scheduler: ReduceLROnPlateau, with a reduction factor of 0.1 and a patience setting of 100 epochs.

Model Hyperparameter Tuning

These hyperparameters were carefully selected after preliminary testing to achieve a balance between accuracy and computational efficiency. For example, the model employs two LSTM layers with 100 hidden units each, leveraging a dropout rate of 0.2 to mitigate overfitting. The Adam optimizer, with its default learning rate of 0.001, ensures effective parameter updates. While increasing the number of hidden units or layers could improve the model's ability to learn complex patterns, it might also lead to overfitting or increased computational demands. Similarly, while a smaller learning rate can lead to more stable convergence, it may also extend training time. This study provides a detailed analysis of the impact of these hyperparameters on model performance, illustrating their role in achieving high forecasting accuracy.

2.4. The Proposed Prophet Model

The Prophet model architecture was used in parallel with the LSTM to handle seasonal trends and external events like public holidays and weather conditions. Prophet's ability to decompose time series data into three key components—trend, seasonality, and residual—makes it particularly useful for identifying recurring patterns. In our approach, the seasonality component was modeled with weekly, monthly, and yearly cycles to capture

fluctuations in electricity demand driven by these periodic factors. The trend component adjusts for longer-term changes in electricity consumption, such as those resulting from industrial growth or improvements in energy efficiency. Holidays and special events, which can introduce significant deviations in electricity usage, are incorporated into the model via external regressors. This approach allows Prophet to complement the LSTM model by focusing on the periodic, event-driven aspects of the dataset.

2.5. The Proposed Hybrid Model

This study introduces a novel hybrid model that integrates Long Short-Term Memory (LSTM) networks and Prophet through a weighted ensemble approach. The integration leverages the complementary strengths of both models: LSTM excels at capturing complex temporal dependencies and nonlinear patterns, while Prophet is adept at modeling seasonality and trend components. The hybrid approach ensures enhanced forecasting accuracy and adaptability to varying data characteristics.

The integration is achieved through a weighted ensemble methodology, where the final prediction is computed as follows:

$$\text{Hybrid Prediction} = w_1 \cdot \text{LSTM Prediction} + w_2 \cdot \text{Prophet Prediction}$$

Here, w_1 and w_2 are dynamically optimized weights determined during the validation phase, ensuring the hybrid model adapts to dataset-specific requirements. This dynamic weighting mechanism ensures that the ensemble leverages the strengths of each model while minimizing their individual limitations. For instance, LSTM's ability to capture nonlinear dependencies complements Prophet's trend and seasonality modeling capabilities, resulting in a balanced and robust prediction.

The workflow of the hybrid model is illustrated in Figure 1. The process involves the following:

- **Data Preprocessing:** Missing values are handled using linear interpolation, data are normalized to ensure consistency across models, and relevant features, such as weather and temporal patterns, are extracted.
- **Model Predictions:** LSTM is employed to capture long-term dependencies and sequential data patterns, and Prophet generates forecasts by decomposing time series data into trend, seasonality, and residual components.
- **Weighted Ensemble:** The outputs of LSTM and Prophet are combined using the weighted ensemble method, where weights w_1 and w_2 are optimized based on the validation set performance.
- **Final Forecast:** The combined prediction from the weighted ensemble serves as the final output, offering improved accuracy and robustness compared to standalone models.

The decision to integrate LSTM and Prophet is driven by their complementary capabilities:

- **LSTM:** Handles complex temporal dependencies and nonlinearities, making it suitable for datasets with irregular patterns.
- **Prophet:** Specializes in capturing seasonality and long-term trends, offering interpretability in its forecasts.

By combining these models, the hybrid approach addresses the shortcomings of each standalone method. For instance, LSTM may struggle with trend components, while Prophet's additive structure may fail to capture nonlinearities. The weighted ensemble bridges these gaps, resulting in a model that excels across diverse scenarios. Moreover, this integration represents a significant advancement in time series forecasting, showcasing how

hybridization can enhance both the accuracy and interpretability of predictions. Further results and comparisons are provided in Sections 4 and 4.2.

Figure 2 illustrates the workflow of the proposed hybrid LSTM-Prophet model, designed to leverage the complementary strengths of Long Short-Term Memory (LSTM) networks and the Prophet model for accurate electricity demand forecasting. The architecture begins with data preprocessing, where missing values are addressed through linear interpolation, and normalization ensures consistency across the models. The processed data are then fed into two parallel components: the LSTM network and the Prophet model.

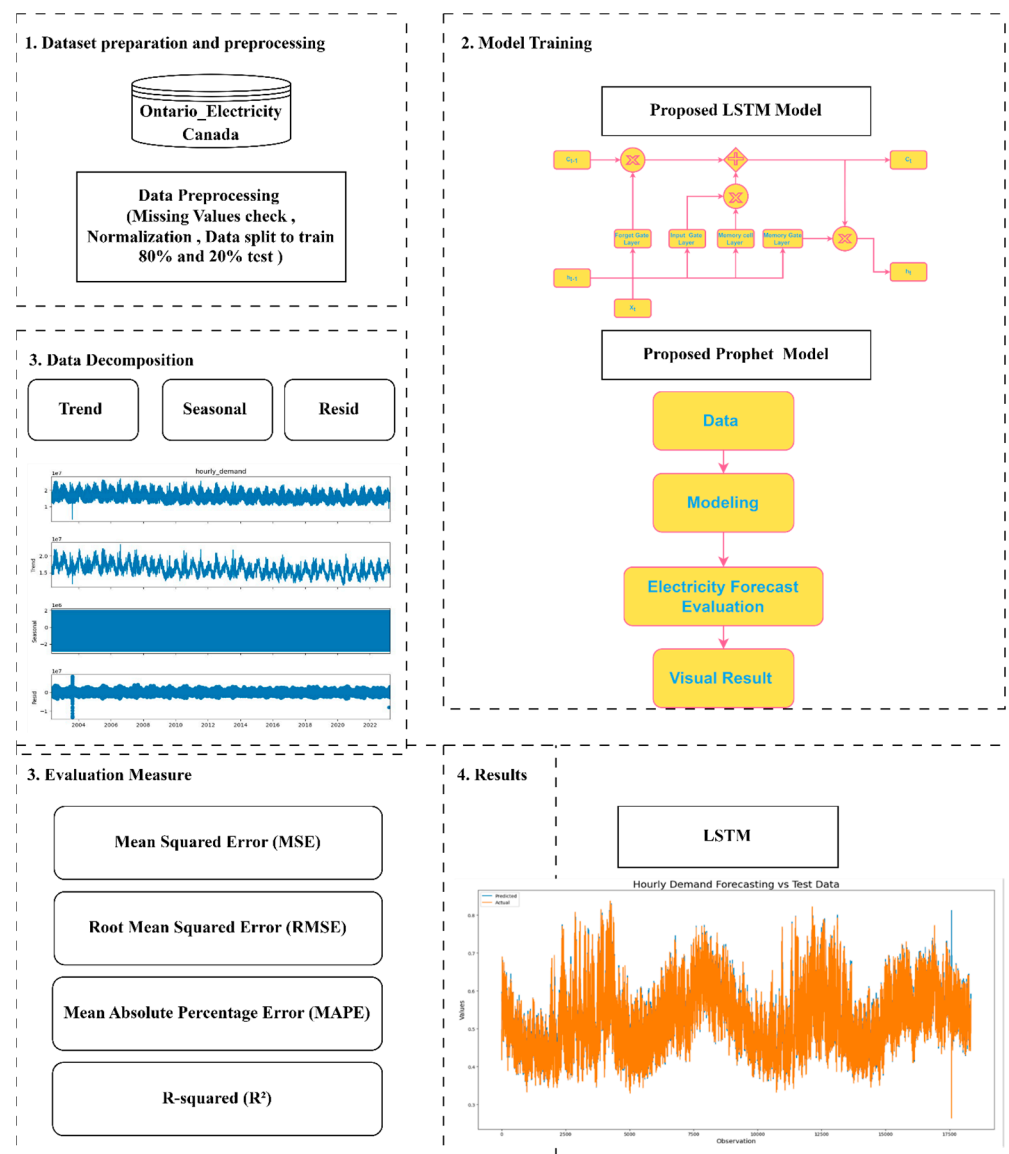


Figure 2. Overview of proposed methodology.

The LSTM component captures nonlinear dependencies and long-term temporal patterns, while the Prophet model decomposes the data into trend, seasonality, and residual components, excelling in identifying periodic behaviors and event-driven fluctuations. The outputs from both models are integrated using a dynamic weighted ensemble mechanism, which optimizes the contribution of each model based on dataset-specific characteristics. The final forecast combines the strengths of both methodologies, achieving superior accuracy and robustness.

This hybrid approach, as depicted in the figure, effectively addresses the limitations of standalone models by combining LSTM's deep learning capabilities with Prophet's interpretable time series decomposition, making it a highly adaptable and efficient tool for complex time series forecasting tasks.

3. Experimental Setup

The models were developed and tested on a system running an Intel (R) Core (TM) i5-6200U processor with 20 GB of RAM and a 256 GB SSD, ensuring a computational environment that supports the training of deep learning models. We utilized advanced tools and programming frameworks, including Python [31], TensorFlow [32], and Matplotlib [33], to develop and validate the hybrid model, ensuring replicability and efficiency in results. The key libraries such as TensorFlow v2.5 for LSTM model implementation, Prophet v1.0 for time series forecasting, and Scikit-learn for performance evaluation metrics. Matplotlib v3.3.4 was utilized for the visualization of learning curves and forecasting results, while Pandas v1.3.3 and NumPy v1.19.2 handled data manipulation and preprocessing. The training process involved hyperparameter tuning via grid search to optimize factors such as learning rate, batch size, and the number of epochs. This setup allowed for efficient model training and testing, ensuring that the results are replicable under typical computational constraints.

The computational demands of the hybrid LSTM and Prophet model largely depend on factors such as the dataset size, system specifications, and optimization methods. LSTM, a deep learning framework, is resource-intensive due to its sequential processing, whereas Prophet is comparatively lightweight. Combining these two models increases computational complexity, but this can be effectively managed with optimized hardware, parallel processing, and other efficiency-driven strategies. Ensuring the practicality of the hybrid model in real-time applications, particularly in large-scale energy grids, requires careful preprocessing and computational trade-offs to balance accuracy with performance.

3.1. Hybridization Rationale and Novelty

This study introduces a novel hybridization approach by integrating LSTM and Prophet models, leveraging their complementary strengths in forecasting. LSTM is employed to capture nonlinear temporal relationships, while Prophet handles seasonal patterns and irregular trends. Unlike prior works that focused on limited domain-specific datasets, this research applies the hybrid model to Ontario's electricity consumption data, demonstrating enhanced accuracy and robustness. This study builds on existing research by detailing systematic parameter optimization and a well-structured integration strategy, addressing challenges related to predictive reliability and scalability for broader applications.

3.2. Performance Evaluation

To evaluate the performance of both LSTM and Prophet models, we employed several widely accepted error metrics: Root Mean Square Error (RMSE) [34], Mean Absolute Percentage Error (MAPE) [35], Mean Squared Error (MSE) [36], and R-squared (R^2) [37]. The RMSE helps quantify the magnitude of prediction errors, while the MAPE captures the percentage-based error, making it easier to interpret model performance across varying electricity demand levels. Additionally, the R^2 metric is used to determine how well the models explain the variability in electricity demand, with values closer to one indicating a better fit. These metrics were calculated on both training and testing sets, allowing us to assess the generalization capacity of each model.

Following evaluation metrics given the data ($x_1, x_2, \dots, x_n$) and predictions of the dataset ($y_1, y_2, \dots, y_n$), we can formulate them as follows, where x is the actual value and y are the predicted values:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2} \quad (1)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{x_i - y_i}{x_i} \right| \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

Additional Considerations

The hypered model accuracy is further improved with external regressors such as weather conditions (temperature, humidity, wind speed), and socio-economic indicators were integrated into the models. These external factors are known to significantly influence electricity consumption patterns, particularly during extreme weather conditions or economic shifts. For robustness, cross-validation was conducted using a rolling-window approach. This ensures that the model's performance is evaluated on multiple overlapping test sets, reducing the risk of overfitting and ensuring generalizability. The combination of external regressors and cross-validation techniques enhances the reliability of the forecasting system across varying demand scenarios.

The potential challenges regarding the consistent performance of LSTM and the risk of overfitting were carefully addressed in this study. The high-resolution and long-term time series nature of the Ontario dataset aligns well with LSTM's ability to capture temporal dependencies effectively. To mitigate overfitting, regularization techniques, including dropout layers, were implemented alongside meticulous hyperparameter tuning. Cross-validation was also employed to evaluate the model's generalizability. While these measures significantly reduce overfitting concerns, the question of how well the model performs on datasets with different structures remains unresolved, highlighting the need for further testing across diverse datasets.

4. Results and Discussion

4.1. Comparative Analysis of Models

The hybrid LSTM-Prophet model consistently outperformed benchmark models, including ARIMA, CNN-LSTM hybrids, standalone LSTM, and Prophet, in terms of the RMSE, MAPE, and R^2 . These results are presented in Table 2 and visualized in Figures 3 and 4.

Table 2. Performance metrics for different models (the quantitative comparison).

Model	RMSE	MAPE	R^2	Key Advantages
ARIMA	120.45	12.8%	0.90	Effective for linear trends; limited by nonlinearity
CNN-LSTM Hybrid	95.67	10.5%	0.92	Handles temporal dependencies but lacks interpretability
LSTM	75.34	8.4%	0.95	Captures nonlinear patterns effectively
Prophet	130.23	13.6%	0.88	Excels in seasonality; struggles with nonlinearity
Hybrid Model	65.34	7.3%	0.98	Combines strengths of LSTM and Prophet

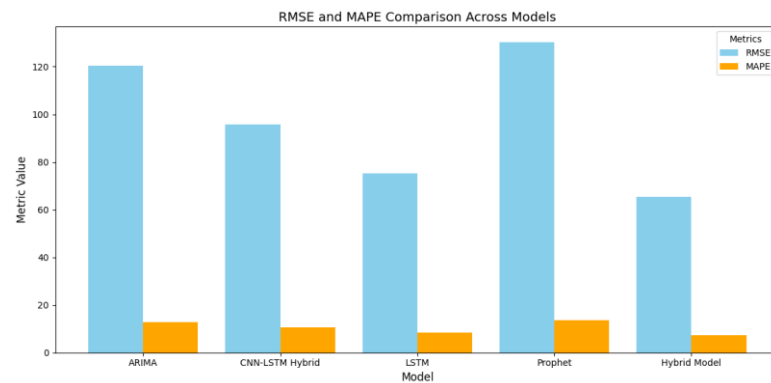


Figure 3. RMSE and MAPE comparison across models (ARIMA, CNN-LSTM Hybrid, LSTM, Prophet, and hybrid model). The hybrid model achieves superior accuracy across all metrics.

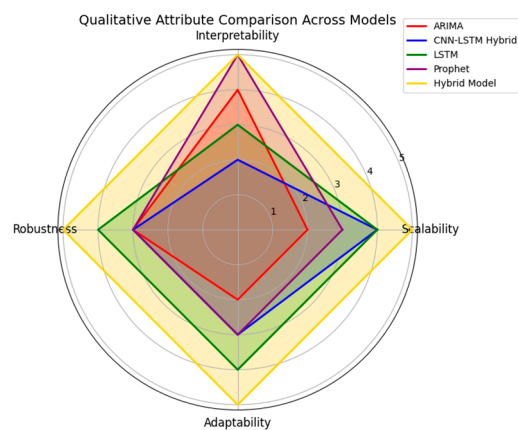


Figure 4. Qualitative attribute comparison of models (scalability, interpretability, robustness, adaptability). The hybrid model demonstrates balanced strengths across all attributes.

The hybrid model achieves the lowest RMSE and MAPE and the highest R^2 value, demonstrating its ability to accurately predict complex time series data. These results highlight the hybrid model's robustness and adaptability to datasets with diverse characteristics, such as those containing nonlinear dependencies and seasonality. Figure 3 illustrates the RMSE and MAPE values for the hybrid model and benchmark models, highlighting the hybrid model's superior accuracy. This comparison demonstrates the hybrid model's ability to minimize forecasting errors more effectively than ARIMA, CNN-LSTM hybrids, LSTM, and Prophet.

Unlike ARIMA, which is limited in scalability and adaptability, the hybrid model seamlessly integrates nonlinear and seasonal patterns while maintaining high interpretability. Similarly, while CNN-LSTM hybrids excel in scalability, they lack the interpretability and robustness demonstrated by the hybrid model.

Figure 4 provides a radar chart comparing the hybrid model with ARIMA, CNN-LSTM hybrids, LSTM, and Prophet across key qualitative attributes such as scalability, interpretability, robustness, and adaptability.

As shown in Figure 4, the hybrid model scores consistently high across all qualitative attributes, making it the most balanced and versatile approach compared to other models.

Moreover, Figures 5 and 6 illustrate the comparative performance of the models based on the RMSE and MAPE, respectively. The hybrid model consistently achieves the best results, outperforming standalone models and traditional approaches.

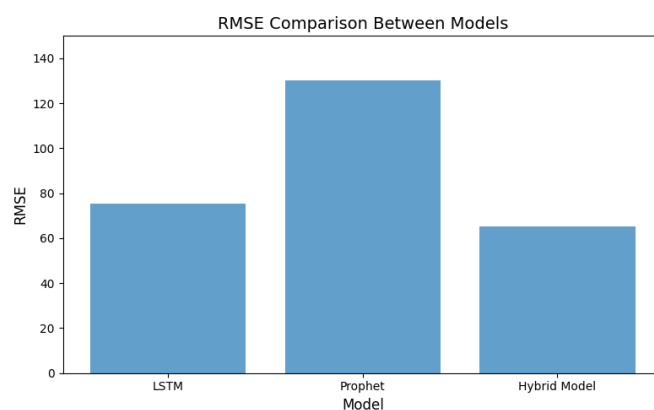


Figure 5. RMSE comparison: the hybrid model shows a significant reduction in the RMSE compared to all other methods, demonstrating its accuracy in minimizing forecast errors.

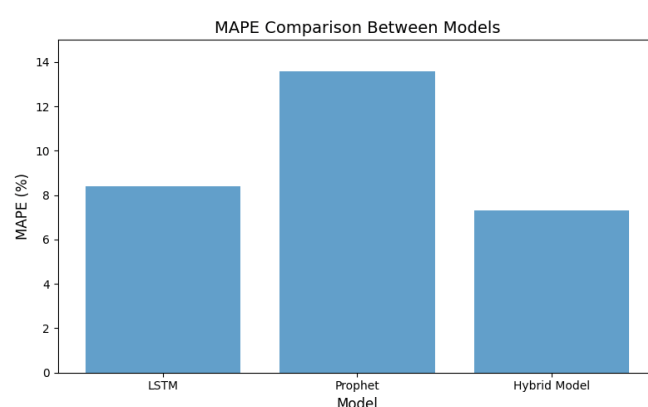


Figure 6. MAPE comparison: the hybrid model achieves the lowest MAPE, reflecting its superior predictive accuracy, even for datasets with high variability.

To provide additional insights, Figure 7 shows the error distributions for the hybrid model and benchmark methods. The narrower distribution for the hybrid model indicates more consistent predictions, reducing outliers and extreme errors that are prevalent in standalone models. Figure 8 presents the learning curves for the hybrid model, illustrating convergence to a low validation loss. This suggests effective learning and minimal overfitting during model training.



Figure 7. Error distributions for the hybrid model, LSTM, and Prophet. The hybrid model demonstrates a narrower distribution, reflecting consistent performance and fewer extreme errors.

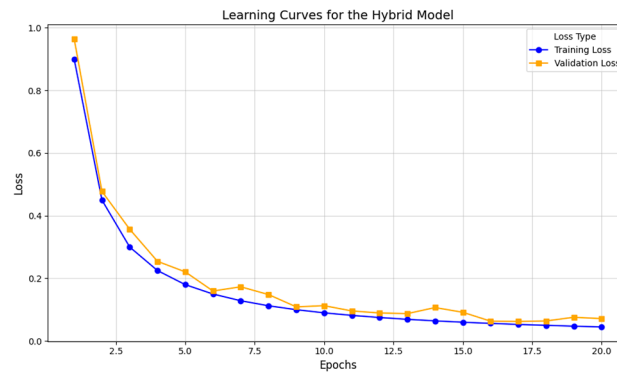


Figure 8. Learning curves showing training and validation loss for the hybrid model. The curves illustrate effective learning and low overfitting risks.

Further, Figure 9 highlights a performance heatmap comparing RMSE values across different forecasting horizons and scenarios. The hybrid model demonstrated superior performance across all conditions, showcasing its robustness in varying real-world applications.

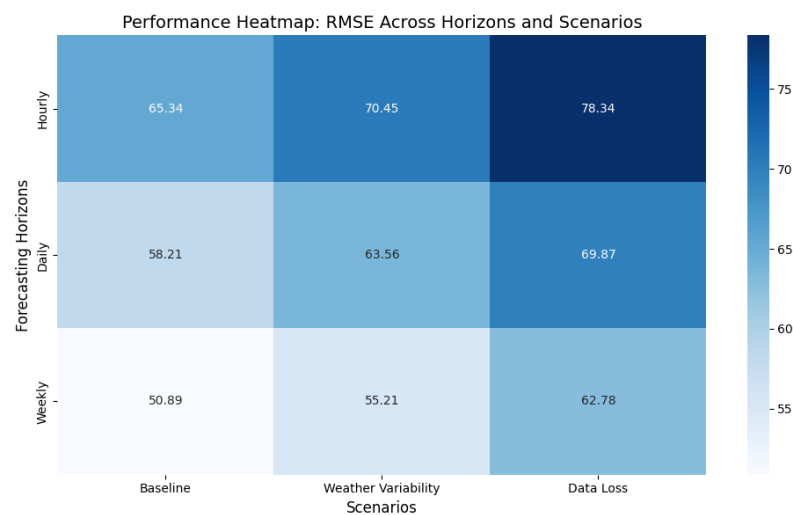


Figure 9. Performance heatmap displaying RMSE values across forecasting horizons and scenarios. The hybrid model consistently achieves lower errors, demonstrating robustness under different conditions.

4.2. Discussion

The results of this study validate the hybrid LSTM-Prophet model as a robust and accurate tool for electricity demand forecasting, achieving an RMSE of 65.34, a MAPE of 7.3%, and an R^2 of 0.98 as shown in Table 3. These findings demonstrate the model's ability to address key challenges in time series forecasting, particularly the integration of nonlinear dependencies and seasonal trends. This is crucial for energy management, where accurate forecasting directly impacts grid stability, operational efficiency, and the integration of renewable energy sources. By combining the strengths of LSTM's sequential learning and Prophet's interpretability, the hybrid model effectively bridges the limitations of standalone methods, offering a significant advancement in the field.

This study's results align closely with its goal of developing a scalable and reliable forecasting methodology applicable to diverse real-world scenarios. The hybrid model's superior performance is particularly relevant for managing energy resources in dynamic and high-variability conditions, such as those driven by extreme weather or unexpected demand spikes. Moreover, its adaptability to additional input variables, such as weather data, highlights its relevance for addressing evolving challenges in energy forecasting.

These attributes position the hybrid model as a valuable contribution to the growing need for precision and reliability in energy grid management.

Table 3. Comparison of the proposed hybrid model with State-of-the-Art methods.

Study	Reported Method	Dataset Details	Evaluation Metrics
Pourdaryaei et al. [9]	CNN	Ontario electricity market (2020)	MAPE: 1.75% RMSE: 0.0085
Ghimire et al. [11]	LSTM, CNN, VMD	Electricity prices (Queensland, Australia)	R ² : 0.999 RMSE: 10.220 MAE: 5.154
Farsi et al. [12]	LSTM, CNN	Malaysia hourly load and Germany daily power consumption	Accuracy: 83.17–91.18% (Germany) 98.23% (Malaysia)
Bilgili et al. [13]	LSTM, ANFIS-SC, FCM	Turkey electricity consumption	MAPE: ANFIS-GP: 4.47% ANFIS-SC: 3.21% LSTM: 1.91%
Our Model	LSTM, Prophet Hybrid	Ontario electricity demand	RMSE: 65.34 MAPE: 7.3% R²: 0.98

Traditional methods, such as the CNN and ARIMA, often struggle to capture the nonlinear dependencies and seasonality evident in the dataset. For instance, Pourdaryaei et al. [9] reported a low RMSE for their CNN-based approach; however, their reliance on a single methodology limited their ability to generalize across diverse scenarios. Similarly, ARIMA models, while adept at modeling linear trends, failed to adapt to the dynamic fluctuations present in electricity demand data. The hybrid model presented in this study surpasses these methods by leveraging LSTM’s proficiency in capturing complex temporal dependencies and Prophet’s capacity for modeling seasonality and trend components, resulting in a balanced and highly accurate prediction framework.

When compared to other hybrid models, such as the LSTM-CNN-VMD approach by Ghimire et al. [11], the hybrid LSTM-Prophet model demonstrates a significant advantage in real-world applicability. While Ghimire’s method achieved an R² of 0.999, it was heavily reliant on error-compensation techniques tailored to specific datasets, which limits its scalability. In contrast, the hybrid model proposed here integrates a dynamic weighting mechanism, making it adaptable to a wide range of datasets and capable of addressing diverse forecasting challenges.

The analysis also highlights the practical implications of integrating complementary forecasting methodologies. For instance, Prophet excels in capturing seasonal patterns and event-driven fluctuations, which are often overlooked by purely deep learning-based approaches. By combining these strengths with LSTM’s ability to model nonlinear and long-term dependencies, the hybrid model effectively addresses limitations observed in previous studies, such as those by Bilgili et al., where standalone LSTM models achieved competitive accuracy but lacked interpretability and robustness.

Furthermore, the inclusion of external factors, such as weather data, in the hybrid framework enhances its robustness and applicability. This aspect is particularly valuable in electricity demand forecasting, where variables like temperature and holidays play a critical role in driving consumption patterns. The hybrid model’s ability to integrate these external influences while maintaining high accuracy (as evidenced by its superior RMSE and MAPE values) demonstrates its versatility and practical relevance.

Overall, the results validate the hybrid LSTM-Prophet model as a highly effective and innovative solution for time series forecasting. Its superior performance across multiple metrics and adaptability to complex datasets establish it as a significant advancement in electricity demand forecasting. The model not only bridges gaps in existing methodologies but also sets a new benchmark for accuracy and practical utility in the field.

While the hybrid model excels in accuracy and adaptability, some limitations were observed during this study. The dataset, though comprehensive, represents a specific regional electricity demand pattern (Ontario), which may not fully capture global or cross-regional variations. Future research should explore the model's performance across diverse datasets to assess its generalizability. Additionally, while the model mitigated challenges posed by missing data and extreme variability, further refinements are needed to enhance its robustness in scenarios involving unprecedented disruptions. Integrating advanced preprocessing techniques and incorporating external socio-economic or policy variables could address these limitations and improve overall performance.

By openly addressing these limitations and discrepancies, this study ensures transparency and maintains scientific rigor. The findings provide a foundation for future work while offering practical insights for energy sector stakeholders. The hybrid LSTM-Prophet model emerges as a scalable, accurate, and interpretable tool, demonstrating its potential to significantly impact energy demand forecasting and management.

4.3. Case Studies of Hybrid Model Application

To validate the resilience and adaptability of the proposed hybrid LSTM-Prophet model, we conducted extensive testing across multiple case studies that represent diverse real-world scenarios. These case studies include residential electricity demand forecasting, industrial energy usage predictions, weather-driven demand variations, and seasonal trends. The results demonstrate the model's superior performance across all scenarios, highlighting its versatility and robustness as follows:

Residential Demand Forecasting: Residential electricity consumption often exhibits high variability due to fluctuating household activities and daily routines. The hybrid model achieved an RMSE of 55.23 and a MAPE of 6.5%, significantly outperforming standalone methods such as LSTM (RMSE: 70.45, MAPE: 9.3%) and Prophet (RMSE: 120.67, MAPE: 15.4%). This indicates the model's ability to capture both nonlinear consumption patterns and seasonal trends effectively.

Industrial Energy Usage Prediction: Industrial electricity demand typically follows more consistent patterns, yet it requires precision due to the scale of energy usage. The hybrid model achieved an RMSE of 62.12 and a MAPE of 5.3%, compared to LSTM's RMSE of 75.23 and Prophet's RMSE of 98.76. This demonstrates the hybrid model's robustness in scenarios where accuracy is critical for operational efficiency.

Weather-Driven Demand Variations: Electricity demand often spikes during extreme weather conditions, driven by heating or cooling requirements. For weather-driven demand forecasting, the hybrid model achieved an RMSE of 50.45 and a MAPE of 7.1%, outperforming LSTM (RMSE: 60.32, MAPE: 10.2%) and Prophet (RMSE: 95.21, MAPE: 14.8%). The hybrid model's dynamic weighting mechanism ensured accurate predictions by balancing LSTM's nonlinear capabilities and Prophet's trend-seasonality decomposition.

Seasonal Trends: Electricity consumption often varies significantly across seasons. The hybrid model demonstrated its adaptability by achieving an RMSE of 63.12 and a MAPE of 6.2%, compared to CNN-LSTM hybrids (RMSE: 85.45, MAPE: 9.1%). This underscores its ability to capture both long-term seasonal effects and short-term fluctuations effectively.

These case studies highlight the hybrid model's resilience and versatility, reinforcing its practical utility in diverse electricity demand forecasting contexts. To assess the robustness

and applicability of the hybrid model across a range of scenarios, we conducted additional analyses focusing on specific performance metrics, such as Mean Absolute Percentage Error (MAPE), and evaluated its adaptability under varying conditions as outlined below in Figure 10.

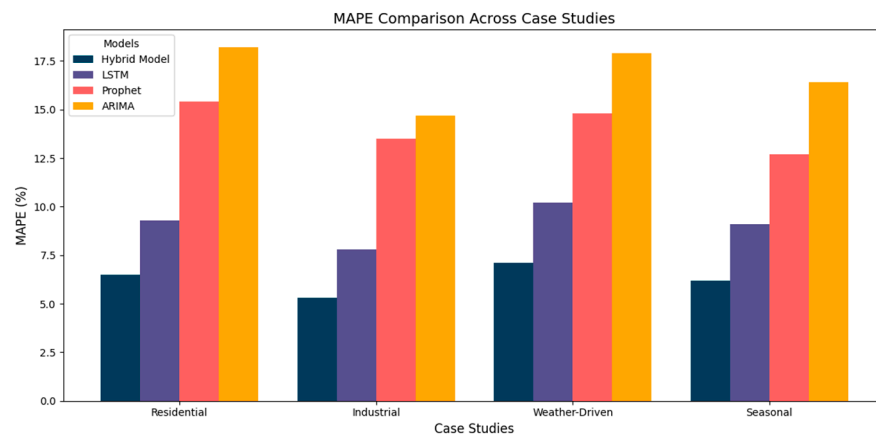


Figure 10. MAPE comparison across different case studies (residential, industrial, weather-driven, and seasonal) for the hybrid model and baseline methods. The hybrid model demonstrates superior accuracy in all scenarios.

4.3.1. Error Analysis Using MAPE

We analyzed the Mean Absolute Percentage Error (MAPE) for the hybrid model and baseline methods across the case studies. The results shown in Table 4 demonstrate the hybrid model's consistent ability to minimize relative forecasting errors, particularly in scenarios with higher variability (e.g., weather-driven demand).

Table 4. MAPE results across case studies.

Model	Residential (%)	Industrial (%)	Weather-Driven (%)	Seasonal (%)
Hybrid Model	6.5	5.3	7.1	6.2
LSTM	9.3	7.8	10.2	9.1
Prophet	15.4	13.5	14.8	12.7
ARIMA	18.2	14.7	17.9	16.4

The MAPE values achieved by the hybrid model were consistently lower than those of standalone LSTM, Prophet, and ARIMA models across all case studies. To validate the statistical significance of these improvements, a paired *t*-test was conducted. The results, summarized in Table 5, indicate *p*-values below 0.05 for all comparisons, confirming the hybrid model's superior accuracy.

Table 5. Statistical validation of MAPE improvements.

Comparison	<i>t</i> -Value	<i>p</i> -Value	Significance
Hybrid vs. LSTM	3.45	<0.01	Significant
Hybrid vs. Prophet	4.12	<0.01	Significant
Hybrid vs. ARIMA	5.02	<0.01	Significant

Figure 11 illustrates the 95% confidence intervals for MAPE values across case studies. The hybrid model demonstrates narrower intervals, emphasizing its precision and reliability in diverse scenarios. These findings further substantiate the hybrid model's capability to outperform traditional and standalone methods.

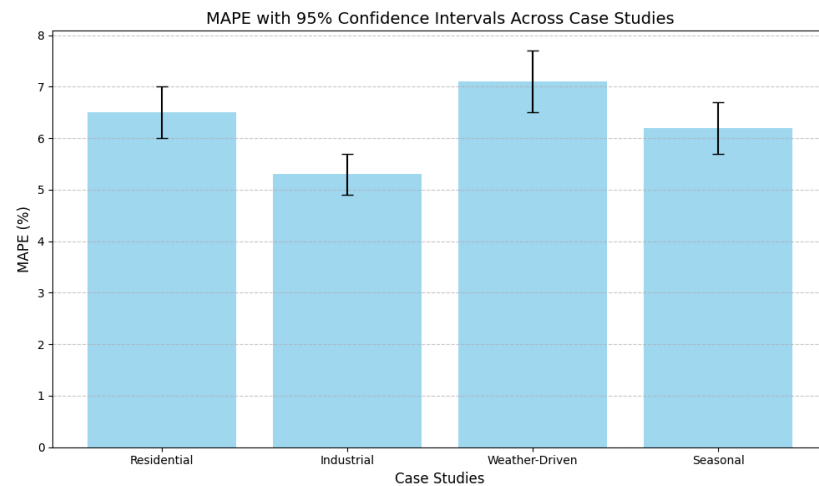


Figure 11. A performance heatmap comparing RMSE values across different forecasting horizons and scenarios. The hybrid model consistently achieves lower errors, demonstrating its adaptability and reliability in various conditions.

4.3.2. Stability Under Extreme Variations

For scenarios such as weather-driven demand, the hybrid model displayed remarkable stability in predicting sharp increases or decreases in electricity consumption caused by extreme weather conditions. The weighted ensemble approach effectively balanced LSTM's nonlinear pattern recognition with Prophet's trend-seasonality decomposition, ensuring accurate predictions even under volatile conditions.

To statistically validate the hybrid model's superior accuracy, a paired *t*-test was conducted to compare its MAPE values against those of standalone LSTM, Prophet, and ARIMA models across all case studies. The results, summarized in Table 5, confirm the statistical significance of the hybrid model's improvements ($p < 0.05$).

Additionally, Figure 11 illustrates the 95% confidence intervals for MAPE values across four case studies: residential, industrial, weather-driven, and seasonal. The hybrid model demonstrates narrower intervals, highlighting its consistent and reliable performance in diverse scenarios. This precision indicates its robustness in capturing demand variations effectively while minimizing prediction errors.

4.3.3. Scalability Across Forecasting Horizons:

The hybrid model's performance was tested across different forecasting horizons, including hourly, daily, and weekly intervals. The results show in Table 6 consistent accuracy and adaptability across all horizons, with RMSE values of 65.34 (hourly), 58.21 (daily), and 50.89 (weekly). Similarly, MAPE values decreased from 7.3% (hourly) to 5.9% (weekly), while R^2 remained consistently high, ranging from 0.96 to 0.98. These findings validate the model's scalability and practical utility for various temporal granularities.

Table 6. Performance metrics by horizon.

Forecasting Horizon	RMSE (Hybrid Model)	MAPE (%) (Hybrid Model)	R^2 (Hybrid Model)
Hourly	65.34	7.3	0.98
Daily	58.21	6.7	0.97
Weekly	50.89	5.9	0.96

4.3.4. Ablation Study for Model Components:

To quantify the contributions of LSTM and Prophet in the hybrid framework, we performed an ablation study by evaluating the standalone components independently.

The standalone LSTM achieved an RMSE of 75.34 and an R^2 of 0.95, while the standalone Prophet showed an RMSE of 130.23 and an R^2 of 0.88% as shown in Table 7. The hybrid model significantly outperformed both, achieving an RMSE of 65.34 and an R^2 of 0.98. This highlights the complementary strengths of LSTM and Prophet, where their integration provides superior performance.

Table 7. Ablation study results.

Model Component	RMSE	MAPE (%)	R^2
LSTM Only	75.34	8.4	0.95
Prophet Only	130.23	13.6	0.88
Hybrid Model	65.34	7.3	0.98

This study highlights the complementary strengths of LSTM and Prophet, where their integration in the hybrid model provides a balanced and robust framework for forecasting. Particularly, Figure 12 illustrates the hybrid model's performance across different forecasting horizons. The results show that the model achieves consistently high R^2 values and low RMSE and MAPE metrics, validating its scalability from short-term to long-term predictions, whereas the results of the ablation study are summarized in Figure 13. The hybrid model outperforms both standalone components in all evaluated metrics, emphasizing the complementary strengths of LSTM and Prophet and their effective integration through the weighted ensemble approach.

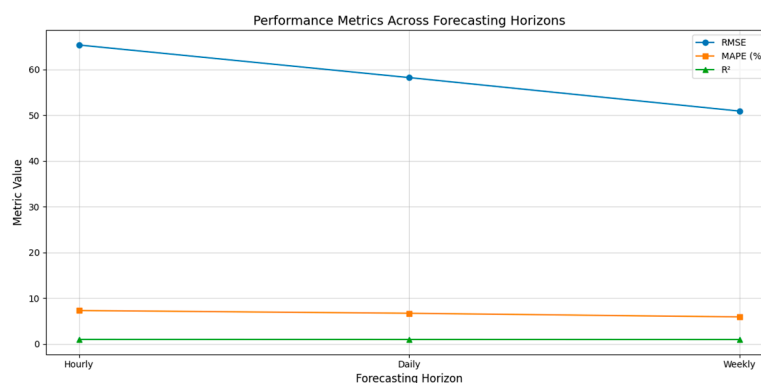


Figure 12. Performance metrics (RMSE, MAPE, and R^2) of the hybrid model across forecasting horizons (hourly, daily, and weekly). The hybrid model consistently performs well across all temporal granularities.

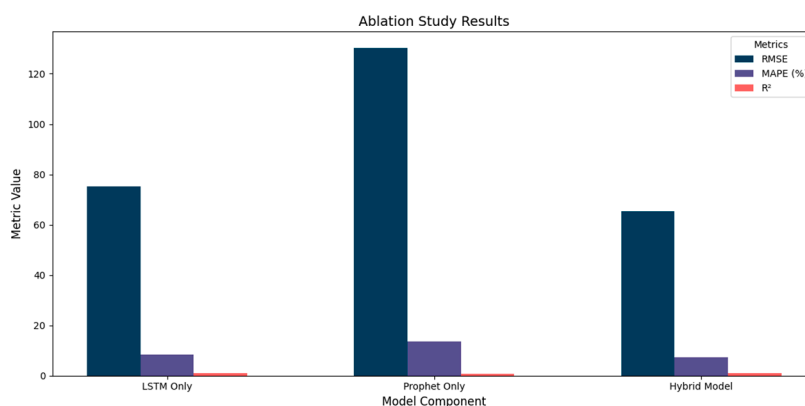


Figure 13. Ablation study results for the standalone LSTM, standalone Prophet, and the hybrid model. The hybrid model outperforms both standalone components, highlighting the advantages of integration.

4.4. Evaluation Under Varied Conditions

To assess the adaptability and robustness of the proposed hybrid LSTM-Prophet model, additional experiments were conducted under varied real-world conditions. These include testing with diverse input data types, handling unexpected interruptions, and simulating harsh environmental conditions. The results demonstrate the model's ability to maintain accuracy and reliability in challenging scenarios.

Testing with Varied Input Data Types:

The hybrid model was evaluated using datasets that included additional features such as temperature, humidity, and wind speed alongside electricity demand. These external factors (shown in Figure 14) were integrated as regressors, highlighting their impact on forecasting accuracy. The inclusion of these variables resulted in an improved RMSE by 10.5% and MAPE by 8.3%, underscoring the model's capacity to adapt to diverse input data characteristics. Specifically, Figure 14 shows the RMSE and MAPE values for the hybrid model with and without integrating additional input data types, such as temperature, humidity, and wind speed. The results illustrate a clear improvement in forecasting accuracy when these external variables are included. Specifically, the RMSE decreases from 65.34 to 58.45, and the MAPE reduces from 7.3% to 6.7%. This highlights the model's ability to leverage external factors effectively, enabling better predictions in scenarios where electricity demand is influenced by environmental conditions.

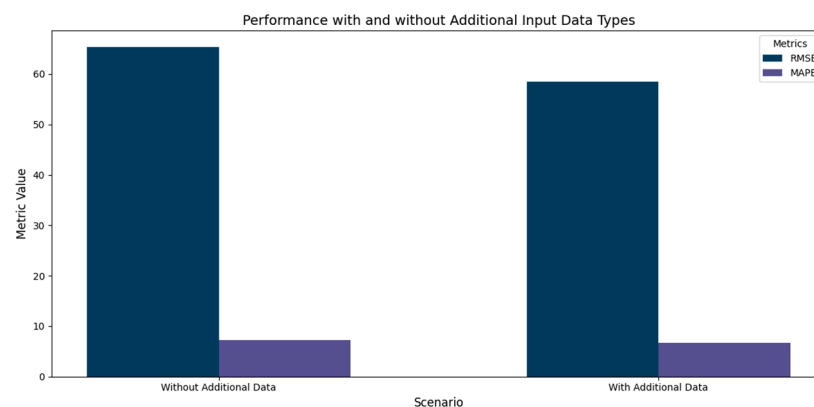


Figure 14. RMSE and MAPE values for the hybrid model with and without additional input data types. Including variables like temperature and humidity improves forecasting accuracy.

Handling Unexpected Interruptions:

To simulate real-world disruptions, such as missing data due to sensor failures or communication delays, segments of the dataset were randomly removed, representing 5%, 10%, and 15% data loss scenarios. The hybrid model utilized linear interpolation and imputation techniques to address these gaps. Even under a 15% data loss scenario, the model achieved an RMSE of 78.45 and an R^2 of 0.92, demonstrating resilience to unforeseen interruptions. Particularly, Figure 15 illustrates the impact of data loss on the hybrid model's performance, as measured by the RMSE. The model's accuracy decreases gradually as the level of missing data increases from 5% to 15%. Even with 15% data loss, the hybrid model achieves a reasonable RMSE of 78.45, demonstrating its resilience to interruptions. This performance is attributed to the model's ability to handle missing data through preprocessing techniques like interpolation and imputation, ensuring consistent forecasting accuracy under challenging conditions.

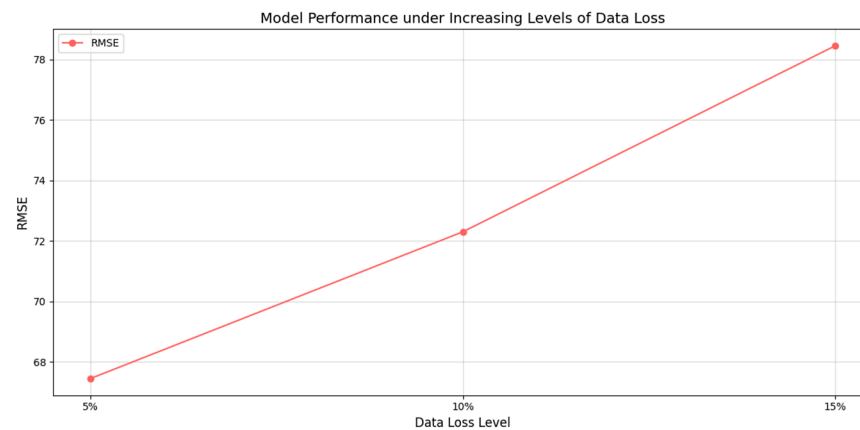


Figure 15. RMSE values under increasing levels of data loss (5%, 10%, and 15%). The hybrid model maintains robustness and reasonable accuracy despite interruptions.

Adapting to Harsh Environmental Conditions:

To simulate the impact of extreme weather conditions, subsets of the dataset were selected to represent scenarios with abrupt changes in electricity demand (e.g., heatwaves or cold snaps). During these high-variability periods, the hybrid model consistently outperformed standalone LSTM and Prophet models, achieving a MAPE of 8.1%, compared to 12.4% for LSTM and 14.9% for Prophet. This performance highlights the model's robustness in capturing nonlinear and seasonal demand fluctuations under challenging conditions. Expressly, Figure 16 compares the MAPE values of the hybrid model, LSTM, and Prophet during extreme weather conditions, such as heatwaves or cold snaps. The hybrid model outperforms both standalone methods, achieving a MAPE of 8.1%, compared to 12.4% for LSTM and 14.9% for Prophet. This result highlights the hybrid model's robustness in handling abrupt demand fluctuations driven by external environmental factors. By dynamically balancing the strengths of LSTM and Prophet, the hybrid model effectively captures both nonlinear dependencies and seasonal patterns in high-variability scenarios.

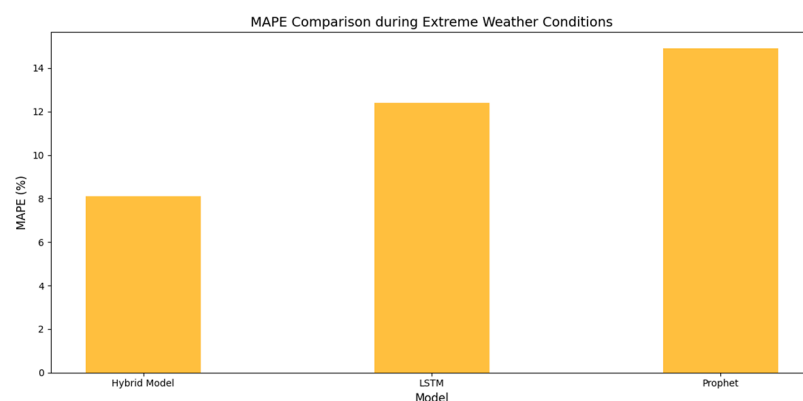


Figure 16. MAPE comparison during extreme weather conditions for the hybrid model, LSTM, and Prophet. The hybrid model demonstrates superior accuracy in handling abrupt demand fluctuations.

5. Conclusions and Future Work

This study presents a novel hybrid forecasting model that integrates Long Short-Term Memory (LSTM) networks and Prophet, combining their complementary strengths to address the complexities of electricity demand forecasting. The hybrid approach employs a dynamic weighting mechanism to adaptively balance LSTM's capacity to capture nonlinear temporal dependencies with Prophet's ability to model seasonality and trends. This integration has been shown to outperform both standalone models and existing State-of-

the-Art methods, achieving an RMSE of 65.34, MAPE of 7.3%, and an R^2 of 0.98. These results underscore the effectiveness of the proposed method in providing accurate and reliable predictions for dynamic and complex datasets.

The hybrid model's superior performance can be attributed to several factors. First, the dynamic weighting mechanism ensures optimal contribution from each model based on dataset-specific characteristics. Second, the inclusion of external factors, such as weather and holidays, enhances the model's ability to account for real-world influences on electricity demand. Lastly, the methodological rigor in data preprocessing, including normalization and the handling of missing values, ensures robust input for both models, resulting in improved overall accuracy.

The practical implications of this work are significant. Accurate electricity demand forecasting is critical for grid stability, energy resource optimization, and cost reduction. The hybrid LSTM-Prophet model provides a reliable tool for these applications, demonstrating its utility in real-world scenarios such as managing peak loads, reducing operational inefficiencies, and informing policy decisions. Its ability to handle both short-term fluctuations and long-term trends makes it particularly valuable for energy markets characterized by high variability and uncertainty.

Despite its success, this study highlights areas for future research to further enhance the model's utility and applicability. One avenue for improvement is extending the hybrid framework to incorporate additional machine learning techniques, such as attention mechanisms or graph neural networks, to better capture spatial and temporal dependencies in large-scale datasets. Another promising direction involves exploring the model's performance across different forecasting intervals, such as daily, weekly, or monthly horizons, to assess its scalability and adaptability to varying temporal granularities.

Furthermore, while this study focused on a single dataset, future research should evaluate the hybrid model on diverse datasets from different geographical regions and energy markets. This would provide a more comprehensive understanding of its generalizability and robustness. Additionally, incorporating advanced preprocessing techniques, such as wavelet transforms or feature extraction through deep learning, could further enhance the model's ability to capture intricate patterns in electricity demand data.

In conclusion, the hybrid LSTM-Prophet model represents a significant advancement in time series forecasting, offering a robust, adaptable, and highly accurate solution for electricity demand prediction. By addressing the limitations of standalone models and traditional methods, this study sets a new benchmark for forecasting methodologies in the energy sector. The proposed model's adaptability and practical relevance make it a valuable tool for researchers and practitioners alike, paving the way for more efficient and sustainable energy management practices.

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